Proceedings of Machine Learning Research [VOLUME # TBD]:1–20, 2026 Machine Learning for Healthcare

Validation Gaps in Drug-Drug Interaction Prediction Benchmarks for Tuberculosis-HIV Pharmacotherapy

Arlene Keith
Whiting School Of Engineering
Johns Hopkins University
Baltimore, MD, US

Abstract

Tuberculosis remains the leading infectious cause of death worldwide, requiring a regimen of four to six drugs over six to nine months. This multi-drug treatment poses a significant risk of drug interactions, which automated tools must detect reliably. Machine learning models are increasingly used to predict these interactions and support clinical decisions, but their validation benchmarks have rarely been audited to assess how well they cover drugs central to global health. This study audits a DDI benchmark built from DrugBank and DDInter using three complementary methods, evaluating seven model variants, including graph neural networks, multilayer perceptrons, and tree-based classifiers, on 130,014 drug pairs. Routine database merging with exact-name matching initially excluded 1,825 documented interactions, including all 284 involving rifampin, which were later recovered by alias resolution. Beyond merging, the partition design was equally restrictive, as 99.9% of test pairs shared both drugs with the training set. This left the model's performance on new agents such as delamanid, pretomanid, and dolutegravir untested. The rare Minor severity class exposed opposing failure modes across architectures, even on familiar drugs. Coverage gaps were not the only source of unreliability. Graph-based models over-flagged cases, while descriptor-based models under-flagged them, a pattern that aggregate metrics mask but that clinical alert design must address. These findings imply that benchmarks constructed this way tend to overestimate the reliability of models for drugs most crucial to global TB and HIV management.

1. Introduction

The standard tuberculosis (TB) treatment combines rifampin, a first-line drug, with several co-medications. Approximately 6.1% of TB patients worldwide also live with HIV and depend on antiretrovirals that share the same metabolic pathway as rifampin (WHO, 2024). Rifampin is a potent CYP3A4 inducer, and when taken with antiretrovirals it can lower their plasma levels enough to cause virological failure (Niemi et al., 2003). The risk runs in both directions, since adverse events from rifampin combined with adjusted-dose lopinavir-ritonavir were severe enough to end a healthy volunteer study (Nijland et al., 2008). These interactions are often predictable from pharmacokinetic profiles, but prediction requires a screening tool the prescriber can access. In resource-limited settings where pharmacist review is limited, automated drug-drug interaction (DDI) tools are increasingly taking on this responsibility.

The machine learning community has developed numerous methods for DDI prediction, ranging from feature-based techniques using molecular fingerprints and pharmacokinetic data to graph neural networks that learn molecular structures. The strong performance of these models on benchmark datasets (Ryu et al., 2018; Nyamabo et al., 2021) has prompted discussions about integrating them into clinical decision support systems. Some tools are now undergoing regulatory review under the FDA's Software as a Medical Device framework, which requires developers to validate safety and effectiveness. Safety and effectiveness are evaluated as aggregate properties of the intended use population rather than as properties of the subgroups within it. Validation for equity, including reliable performance across diverse populations, largely falls outside this framework, and the resulting oversight lands hardest on groups whose medications are underrepresented in the source data. The reported benchmark results also depend on specific assembly choices that have not been scrutinized, yet these choices significantly influence which drugs the models can accurately predict.

This study evaluates a drug-drug interaction (DDI) prediction benchmark derived from DrugBank and DDInter, utilizing standard methods from existing literature. It continues recent efforts to critique clinical machine learning benchmarks (Wong et al., 2021; Kapoor and Narayanan, 2023; Shen et al., 2025), focusing specifically on the DDI domain. The impact of these methodological choices is particularly significant in cases like tuberculosis-HIV polypharmacy. The treatment involves a limited set of drugs known by WHO nonproprietary names (INN), which closely overlap with antiretrovirals through shared metabolic pathways. The analysis identifies three methodological patterns that have clinical implications. Standard cross-database merging often results in name-resolution gaps for drugs labeled with WHO INN, a known issue when combining databases with different naming conventions. Random pair splits saturate the drug space, making cold-start evaluation structurally impossible. Reporting aggregate metrics can hide architecture-specific failures, especially in rare severity classes. Since drugs identified by WHO INN are most affected by these patterns, the audit holds particular relevance for tuberculosis and HIV treatments. The seven model variants mainly serve as tools to detect these patterns rather than stand as standalone contributions. A multilayer perceptron using fingerprint and pharmacokinetic features outperforms a graph neural network by 9.3 accuracy points overall. In the rare Minor severity class, the two architectures perform poorly but in opposite directions, highlighting how ranking results can mask meaningful differences in alert behaviors that have direct clinical implications.

Generalizable Insights about Machine Learning in the Context of Healthcare

Several common practices in assembling clinical machine learning benchmarks can systematically disadvantage drugs prioritized for global health. The exact-name matching silently drops interactions for drugs identified by their WHO INN names rather than American generic names, which are especially important in TB and HIV treatment. Random pair-level splits with extensive negative sampling saturate the drug space and make cold-start evaluation structurally difficult, despite the fact that newer agents such as delamanid and pretomanid require the strongest evidence. Additionally, aggregate accuracy and macro-F1 can mask architecture-specific errors in rare-severity classes, where models with comparable overall scores may over-alert or under-alert in opposing ways, creating patient safety risks that fall unevenly across populations. These concerns are not exclusive to DDI prediction. Any clinical benchmark that integrates cross-database curation with random pair splits encounters similar risks, and validation gaps in these benchmarks may not be detectable through standard reporting.

2. Related Work

Machine learning methods for DDI prediction mainly fall into two categories. Feature-based approaches rely on precomputed molecular descriptors, while representation-learning methods create embeddings directly from molecular structures. For example, Ryu et al. (2018) developed DeepDDI, a deep neural network that classifies pairs into 86 DDI types based on structural similarity profiles. Zitnik et al. (2018) used a different strategy by applying a graph convolutional network on a multi-relational graph of drugs and protein targets to predict polypharmacy side effects. Recent advancements include substructure-attention mechanisms (Nyamabo et al., 2021) and contrastive representation learning (Wang et al., 2021), with several of these systems now being considered for integration into clinical decision support.

A parallel line of research has started questioning whether the benchmark results of clinical machine learning models accurately predict their real-world performance. Nestor et al. (2019) demonstrated that the robustness of features in non-stationary health records significantly affects model performance after deployment. Kapoor and Narayanan (2023) expanded on this concern by showing how data leakage can artificially boost performance metrics across various machine learning fields. The gap between development and real-world application is real and evident. Wong et al. (2021) reported that a widely used sepsis prediction model performed much worse in real-world testing than in benchmarks. Huang et al. (2021) responded by establishing the Therapeutic Data Commons, aiming to standardize evaluation methods for biomedical machine learning and reveal how different data partitioning strategies impact reported results.

In drug-drug interactions (DDI), this deployment gap manifests as alert fatigue. Felisberto et al. (2024) observed that prescribers dismiss alerts more quickly than they read them, with override rate emerging as a key failure point in clinical decision support systems. This issue heavily burdens prescribers managing high-volume polypharmacy populations, where TB-HIV coinfection is common globally. Models that over-alert on frequent interactions are not only metrically suboptimal but actively harmful in practice, as dismissed alerts accumulate and erode prescriber trust in the system.

Recent DDI research has started to converge on similar themes. Liu et al. (2022) introduced CSMDDI for cold-start DDI prediction, bridging the gap between random-split benchmarks and real-world scenarios involving new drugs. Joeres et al. (2025) created DataSAIL to produce biomedical data splits with minimal leakage, while Shen et al. (2025) presented DDI-Ben, a framework that evaluates ten DDI methods under simulated distribution shifts, revealing that many perform poorly outside random splits, especially with drug-disjoint data. None of these studies specifically address how choices in benchmark assembly influence the drugs that anchor global tuberculosis and HIV treatment. This study aims to fill that gap by examining drugs listed under the WHO international nonproprietary nomenclature that are most susceptible to these methodological variations.

3. Methods

This work evaluates seven model variants grouped into three families. Each variant is tested on the same held-out dataset using a standardized probability-native pipeline (Section 5.1), ensuring that the differences in Section 5 are due to the models rather than scoring artifacts. While the model designs are not new, the focus is on their evaluation methods and the insights these results provide about benchmark validity. This section details their architectures, loss functions, training procedures, and the evaluation setup. Section 4 explains the data pipeline that supplies these models. During pipeline setup, an indirect label leakage in the pharmacokinetic features was identified and corrected before training; the methodology is documented in Section 4 for researchers aiming to reproduce similar benchmarks. All numerical results in Section 5 are obtained from models trained on the leakage-corrected pharmacokinetic vector.

3.1. Model Variants

The seven model variants include three baseline classifiers that depend on fixed molecular descriptors, three simplified versions of the full graph neural network (GNN) architecture, and one comprehensive multi-task GNN. These baselines, together with the full GNN+PK, present different architectural perspectives on the same drug pairs, making class-stratified disagreement a main aspect of the third audit finding. The ablation studies help identify which input streams provide signals that each model prioritizes and clarify whether multi-task training genuinely enhances severity prediction or whether the mechanism head mainly relies on the residual signal in the leakage-corrected pharmacokinetic vector, as discussed in Section 4.

The baseline models use a concatenated feature vector combining Morgan fingerprints and pharmacokinetic descriptors. For each drug pair, the input vector is structured as

$$[\mathbf{fp}_a;\ \mathbf{fp}_b;\ \mathbf{fp}_a \odot \mathbf{fp}_b;\ |\mathbf{fp}_a - \mathbf{fp}_b|;\ \mathbf{pk}_a;\ \mathbf{pk}_b] \tag{1}$$

where $\mathbf{fp}$ is a 2,048-bit Morgan fingerprint at radius 2 and $\mathbf{pk}$ is a 10-dimensional pharmacokinetic descriptor vector with leakage correction. This yields 8,212 features per pair. The three classifiers, random forest, multilayer perceptron, and XGBoost (RF, MLP, and XGB), are trained separately for severity and mechanism predictions because they are not jointly optimized. To ensure well-calibrated probabilities, the RF and XGB models use `CalibratedClassifierCV` with three-fold isotonic regression, addressing overconfidence issues common with uncalibrated tree ensembles compared to GNN softmax outputs. Hyperparameters are detailed in Appendix A.

The comprehensive multi-task model (GNN+PK) integrates a graph neural network encoder for molecular structures with a distinct branch for pharmacokinetics. The molecular encoder employs three GATv2 layers (Brody et al., 2022) with edge attributes, incorporating jumping knowledge max-pooling (Xu et al., 2018) across layers and a mean-max readout, resulting in a 256-dimensional embedding per molecule. The pharmacokinetic branch is a two-layer MLP transforming a 10-dimensional descriptor vector into a 64-dimensional embedding. These embeddings are combined into a 320-dimensional per-drug representation $\mathbf{h}$. For a drug pair, the interaction vector is constructed as

$$[\mathbf{h}_a;\ \mathbf{h}_b;\ \mathbf{h}_a \odot \mathbf{h}_b;\ |\mathbf{h}_a - \mathbf{h}_b|] \tag{2}$$

which is then processed through a two-layer interaction trunk with two task-specific heads for severity and mechanism predictions. Three ablation variants are employed to analyze the components of the complete model. The GNN-only version omits the pharmacokinetic branch, relying solely on molecular embeddings while maintaining both task heads. The PK-only variant removes the graph encoder, depending entirely on the pharmacokinetic embedding. The single-task variant preserves the full architecture but sets the mechanism loss weight to zero during training, isolating the effect of multi-task learning on severity prediction.

Table 1: Seven model variants evaluated on a shared pipeline. Three baselines and the full GNN+PK use both molecular and pharmacokinetic inputs. Three ablations isolate components by removing one input or the mechanism task.

Model	**Family**	**Molecular input**	**PK input**	**Severity**	**Mechanism**
RF	Baseline	Morgan FP	✓	✓	✓
MLP	Baseline	Morgan FP	✓	✓	✓
XGB	Baseline	Morgan FP	✓	✓	✓
GNN+PK	Full	GATv2 graph	✓	✓	✓
GNN-only	Ablation	GATv2 graph		✓	✓
PK-only	Ablation		✓	✓	✓
Single-task	Ablation	GATv2 graph	✓	✓	

3.2. Loss and Training

The data are divided into training, validation, and test sets with an 80/10/10 split, stratified by severity class to preserve the natural distribution across subsets. Negative pairs are sampled at a 1:1 ratio relative to documented positives, selected from non-interacting drug combinations in the DrugBank pool. All seven model variants use the same data split and negative pool, ensuring that any comparison between models is made using identical pairs.

Training optimizes a weighted sum of severity and mechanism losses,

$$\mathcal{L} = \mathcal{L}_{\text{sev}} + \lambda\, \mathcal{L}_{\text{mech}} \tag{3}$$

The severity loss employs class-weighted cross-entropy with inverse-frequency weights calculated on the training split. The mechanism loss uses binary cross-entropy with logit positive weights computed as $\log(1 + n_{\text{neg}}/n_{\text{pos}})$. The log-dampened positive weight stabilizes training for rare mechanism classes such as QT prolongation and protein binding, which were unstable under standard inverse-frequency weighting in pilot runs. The mechanism weight λ is set to 0.5 in the full multi-task model and to 0 in the single-task ablation. The optimizer is Adam with a learning rate of 10^{-3} and a weight decay of 10^{-5}. A cosine annealing schedule decays the learning rate to $\eta_{\min} = 10^{-6}$ over a maximum of 80 epochs. Training uses early stopping with patience 15 and gradient clipping at norm 1.0. The batch size is 512.

The early stopping criterion is based on the severity validation loss across all four GNN variants rather than each variant's training objective. This separates the model selection process from the training goal. Without this separation, the single-task variant would stop training solely based on severity loss while the multi-task variants would stop based on the joint loss, making direct comparisons unfair. Model training was conducted on an Apple M4 Max with 128 GB RAM using the Metal Performance Shaders (MPS) backend for GPU acceleration. While random seeds are set at the start of each run, the MPS backend introduces slight non-determinism, resulting in metric differences across reruns that fall below the comparison resolution discussed in Section 5.

3.3. Evaluation Framework

All seven model variants are evaluated using a common test set of 13,002 drug pairs from the alias-corrected merge described in Section 4. They use a unified scoring pipeline that accounts for predicted probabilities. Severity classification is scored on accuracy, macro F1, macro AUROC, and macro AUPRC. For mechanism classification, outcomes are reported as macro-averaged AUROC across mechanisms, using both positive and negative test examples. Classification thresholds are optimized on the validation set by maximizing F1 before being applied to the test set. F1 balances precision and recall in the face of significant class imbalance, with many labels appearing in fewer than 1% of pairs. Calibration accuracy is evaluated using the expected calibration error (ECE), calculated over 10 equally spaced bins. The selection and calibration of thresholds have direct clinical impacts, as miscalibrated probabilities and poorly chosen thresholds can lead to unreliable alert levels for prescribers and uneven safety margins across patient groups.

Pairwise model comparisons combine McNemar's test on paired severity correctness with a paired bootstrap of 1,000 resamples to construct 95% percentile intervals for the accuracy difference. The same bootstrap procedure is applied to construct confidence intervals for individual metrics including accuracy, macro-F1, severity AUROC, and mechanism AUROC.

The test pairs are categorized based on whether the drugs belong to the training set, aligning with the approach described in the second audit finding. The partition definitions appear in Table 3. The class-stratified disagreement analysis, which addresses the third audit finding, focuses on pairs in which two models yield different severity predictions. For each true severity class, it reports the number of pairs where each model is correct independently, emphasizing class-specific differences in model performance that aggregate accuracy metrics may obscure.

For the baseline models, 5-fold stratified cross-validation is performed on the training data to ensure hyperparameter stability. Models from each fold are saved separately and tested on the same validation and test sets. Results are reported as means and standard deviations across folds. A 5-fold ensemble that averages predicted probabilities across folds is also evaluated to compare directly with the single-checkpoint GNN. Probabilities are averaged separately for severity using a softmax function and for mechanism classification using sigmoid outputs.

4. Cohort

This work uses two publicly accessible drug interaction databases. DrugBank (version 5.1.15) provides drug structures, pharmacokinetic annotations, and narrative descriptions of interactions (Wishart et al., 2018). DDInter database provides interaction pairs categorized into three severity levels (Xiong et al., 2022). The merge produces a single set of drug pairs, each with a severity label, a binary mechanism vector from the interaction text, and feature representations for both drugs. A fourth severity class, namely "none," is added through negative sampling described in Section 4.1. These databases mainly rely on regulatory filings and pharmacology literature from high-income countries, influencing which drug pairs are documented and which are excluded. This asymmetry forms the basis of the first audit finding discussed in Section 5.

4.1. Cohort Selection

Drug pairs were included only if both molecules had valid SMILES strings in DrugBank and complete pharmacokinetic data. After applying these criteria, the dataset comprised 65,007 positive pairs, each matched with an equal number of sampled negatives, totaling 130,014 pairs. Negative pairs were chosen from valid drug combinations not listed in DrugBank's interaction database and were labeled "none" for severity, distinguishing them from the three documented severity levels (Minor, Moderate, Major). The data were split into 104,011 training pairs (80%), 13,001 validation pairs (10%), and 13,002 test pairs (10%). All seven model variants use these same splits and negative samples, allowing for consistent cross-model comparisons.

Treating unlabeled pairs as negatives aligns with the positive-unlabeled assumption, which should be explicitly acknowledged. Because DrugBank records only known interactions, any pair lacking a recorded interaction is considered non-interacting during training. This may result in overestimating specificity and underestimating false negatives, particularly for less-studied drug pairs where missing data could be mistaken for no interaction. The negative-sampling approach reduces but does not fully eliminate this bias.

The integration of DDInter and DrugBank revealed a name-resolution gap rather than a seamless merge. DDInter adheres to WHO INN standards, while DrugBank uses American generic names. Two examples illustrate the asymmetry. DDInter records the drug as "rifampicin" while DrugBank uses "rifampin," and DDInter records "salbutamol" while DrugBank uses "albuterol." An exact-string merge initially removed 1,825 interactions, approximately 1.5% of DDInter records, including all 284 pairs involving rifampin, a crucial CYP3A4 inducer used to treat tuberculosis. An analysis of unmatched DDInter names revealed three recoverable mismatches addressable through an alias map, involving 1,104 pairs for albuterol, 284 for rifampin, and 231 for interferon alfa-2a variants. After applying alias corrections, 197 pairs still lacked a definitive DrugBank match and were excluded. The bias is asymmetric, with drugs listed under WHO INN, commonly used in tuberculosis and HIV guidelines worldwide, most likely to be omitted. Section 5.2 revisits this issue with the case of rifampin and CYP induction. The remainder of the paper uses "rifampin" for the drug itself.

The TB cohort consists of 18 drugs that passed the alias-corrected merge, each with at least 10 test pairs, categorized into four clinical tiers representing standard TB and TB-HIV treatments. This structure reflects rifampin's role in inducing cytochrome P450 enzymes through the pregnane X receptor pathway, which explains most interactions across tiers (Niemi et al., 2003; Burman et al., 2001). The first-line regimen includes rifampin, isoniazid, and ethambutol, forming the core treatment for drug-sensitive TB. Second-line oral agents such as bedaquiline, linezolid, moxifloxacin, levofloxacin, clofazimine, and amikacin are used when initial therapy fails or resistance occurs. Common co-administered antiretroviral drugs in HIV-positive TB patients include efavirenz, ritonavir, nevirapine, atazanavir, and lopinavir, selected for their compatibility with rifampin's enzyme induction. Additional medications such as fluconazole, voriconazole, metformin, and glipizide are noteworthy for their interactions during treatment for opportunistic infections and metabolic complications. Drugs including pyrazinamide, delamanid, pretomanid, dolutegravir, ethionamide, cycloserine, tenofovir, and emtricitabine are excluded due to insufficient test coverage, a consequence of the second audit finding discussed in Section 5.3.

4.2. Data Extraction

DrugBank offers three data categories, each requiring distinct extraction approaches. Molecular structures are provided as SMILES strings, which RDKit transforms into molecular graphs for GNN use. Drugs with invalid or absent SMILES are omitted from the cohort. Pharmacokinetic annotations detail how each drug interacts with metabolic enzymes and transporters. The cytochrome P450 family covers eight enzymes (3A4, 2D6, 2C19, 2C9, 1A2, 2B6, 2E1, and 2C8), each annotated with three possible roles (substrate, inhibitor, and inducer). Two transporters share the same three-role structure, with P-glycoprotein and OATP1B1 selected for their relevance to co-administration of TB and HIV drugs. Each drug receives a 30-bit binary annotation vector at extraction. An earlier extraction pass searched for the substring "P-glycoprotein 1," which does not appear in DrugBank's transporter records under that name. The actual label is "ATP-dependent translocase ABCB1," which went unmatched, and 608 drugs, including rifampin, lost their P-gp annotation. The corrected query repopulates these flags before any downstream feature engineering.

Interaction mechanisms in DrugBank are presented as free-text descriptions rather than structured data fields. These labels were assigned through regex pattern matching across seven categories covering CYP induction, CYP inhibition, QT prolongation, additive toxicity, absorption interference, protein binding displacement, and renal excretion. A drug pair is considered positive for a mechanism if its interaction description matches the relevant pattern. Approximately 19.1% of positive pairs do not match any keyword, forming an unlabeled group the classifier does not learn from. Variations in terminology within DrugBank descriptions can introduce noise in the labels, which the model may interpret as systematic error rather than genuine biological uncertainty. This limitation is addressed further in Section 6.

4.3. Feature Choices

Three feature representations are created from the data, one for each model group. The GNN variants analyze a molecular graph where each atom has a 49-dimensional feature vector. Most dimensions encode categorical chemical properties using one-hot encoding. The atomic number includes nine common organic elements (carbon, nitrogen, oxygen, fluorine, phosphorus, sulfur, chlorine, bromine, and iodine) plus a miscellaneous category.

Additional features include total degree, formal charge, hybridization, hydrogen count, and explicit valence, all encoded with one-hot vectors. Six scalar properties are added, covering aromaticity, ring membership, heteroatom status, minimum ring size, atomic mass, and radical electron count. Bonds are described by a 14-dimensional feature vector with one-hot encoding for bond type and stereo configuration, plus two binary flags for conjugation and ring participation. This featurization matches standard practices in graph-based molecular property prediction (Brody et al., 2022; Nyamabo et al., 2022).

The baseline classifiers use Morgan fingerprints (Morgan, 1965; Rogers and Hahn, 2010). Each drug's SMILES is transformed into a 2,048-bit fingerprint at radius 2 through RDKit. The input vector for each pair includes fingerprint $\mathbf{fp}_a$, fingerprint $\mathbf{fp}_b$, their element-wise product, the absolute difference, and both pharmacokinetic vectors, aggregating to 8,212 features after leakage correction. Interaction features are captured through the product and absolute difference terms, following the format

$$[\mathbf{h}_a;\ \mathbf{h}_b;\ \mathbf{h}_a \odot \mathbf{h}_b;\ |\mathbf{h}_a - \mathbf{h}_b|] \tag{4}$$

used in the GNN pair classifier. This ensures that both the baseline and GNN receive comparable pair representations. The main difference is whether the per-drug embedding is fixed (Morgan FP) or learned (GATv2 graph encoder).

The pharmacokinetic (PK) data was audited and corrected after per-mechanism contingency analysis on the training pairs showed that some PK columns contained downstream information from the same DrugBank curation process that generates mechanism labels. Pairs with a CYP inducer flag were 2.5 times more likely to carry a CYP induction label, and those with a CYP inhibitor flag were 2.2 times more likely to carry a CYP inhibition label. These ratios indicate that models with direct access to these flags can infer the mechanism from input features rather than from molecular structure. To prevent this shortcut, 20 columns were removed from the original 30-bit vector, including all CYP-inducer and CYP-inhibitor flags (16 columns), transporter-inhibitor flags (2 columns), and transporter-substrate flags (2 columns). The remaining 10 columns include CYP-substrate flags for all eight enzymes and inducer flags for two transporters, requiring the model to use structural reasoning to predict mechanisms. All metrics in Section 5 are based on models trained and evaluated with this 10-dimensional, leakage-corrected vector.

The identical 10-dimensional pharmacokinetic vector is provided to all models that use PK input. Baseline classifiers incorporate it through concatenation. The GNN+PK variant transforms it into a 64-dimensional embedding using a two-layer MLP before merging it with the molecular embedding. Using this shared vector across model types ensures that performance differences are attributable to representational processing rather than variation in the underlying data.

5. Results

5.1. Evaluation Approach

Seven model variants were assessed on a single held-out test set of 13,002 drug pairs using the metrics described in Section 3.3. Confidence intervals are obtained from 1,000 bootstrap resamples with 95% percentile bounds, and pairwise model comparisons combine

McNemar's test on paired severity correctness with paired bootstrap intervals on accuracy differences.

The remaining subsections correspond with the three main findings of the audit rather than showing isolated metrics. Section 5.2 examines how the cross-database name-resolution issue affects results, using the rifampin CYP-induction case as an example. Section 5.3 assesses partition saturation through random pair-level splits, emphasizing what the test set can and cannot verify. Section 5.4 investigates disagreements between architectures within specific severity classes and the clinical alert behaviors these disagreements imply. Section 5.5 focuses on performance within the TB cohort, where all three findings converge. The headline metrics for the full test set are presented in Table 2 to provide context for the following subsections.

Table 2: Headline metrics on the fixed test set (n = 13,002). Baseline rows show 5-fold cross-validation ensemble means with bootstrap 95% confidence intervals. GNN rows present results from a single run using the best validation checkpoint trained on the leakage-corrected pharmacokinetic vector.

Model	**Accuracy**	**Macro-F1**	**Sev AUROC**	**Mech AUROC**	**ECE**
RF (5-fold ensemble)	0.849 [0.843, 0.855]	0.707 [0.689, 0.722]	0.959 [0.956, 0.962]	0.960 [0.949, 0.968]	0.017
MLP (5-fold ensemble)	0.955 [0.951, 0.958]	0.896 [0.884, 0.907]	0.991 [0.989, 0.992]	0.984 [0.975, 0.990]	0.011
XGB (5-fold ensemble)	0.890 [0.885, 0.895]	0.781 [0.766, 0.795]	0.973 [0.971, 0.976]	0.974 [0.960, 0.982]	0.014
GNN+PK (full)	0.862 [0.856, 0.868]	0.748 [0.736, 0.759]	0.973 [0.971, 0.976]	0.948 [0.941, 0.955]	0.008
GNN-only	0.849 [0.842, 0.855]	0.742 [0.730, 0.754]	0.970 [0.968, 0.973]	0.947 [0.939, 0.956]	0.010
PK-only	0.635 [0.627, 0.643]	0.472 [0.461, 0.482]	0.828 [0.821, 0.835]	0.823 [0.806, 0.840]	0.106
Single-task	0.851 [0.845, 0.857]	0.727 [0.715, 0.739]	0.971 [0.969, 0.974]	N/A	0.009

The MLP ensemble outperforms other models in accuracy, macro-F1, and severity AUROC. GNN+PK has the lowest expected calibration error at 0.008, while the other main models range between 0.010 and 0.017. The PK-only ablation results in 0.635 accuracy on the leakage-corrected vector, suggesting that the 10-dimensional pharmacokinetic representation alone is insufficient and that the primary signal for GNN+PK comes from the molecular branch. The accuracy of the single-task ablation matches that of the multi-task GNN+PK within bootstrap variability, indicating that the mechanism-prediction objective contributes little to severity accuracy in the leakage-corrected setting.

5.2. Name-Resolution Failure and Global-Health Drug Coverage

The 16 rifampin pairs that passed the alias-corrected merge enable targeted assessment of CYP induction prediction, the most important interaction in tuberculosis treatment. The GNN+PK model predicts CYP induction for these pairs with perfect accuracy, correctly identifying all 14 true positives and avoiding false positives and negatives, yielding an F1 score of 1.000. The MLP, XGBoost, and Random Forest ensembles each achieve an F1 score of 0.933, with a recall of 1.000 and a precision of 0.875.

These results could not be achieved with any of the seven models without the alias correction. An exact-string DDInter-DrugBank merge removes all rifampin pairs, excluding the most clinically significant CYP-induction case in the TB-HIV regimen from the benchmark before any model has a chance to evaluate it. The method chosen for name resolution during benchmark setup determines which clinical questions the models can answer. In this case, whether a model correctly flags rifampin-driven CYP induction only became answerable after identifying and correcting the merge gap during the audit.

5.3. Partition Saturation and Cold-Start Impossibility

The random pair-level splits with broad-pool negative sampling generate a training set that nearly encompasses the entire drug space. Out of 13,002 test pairs, both drugs are present in the training set in 12,997 cases (99.96%). Only 5 test pairs include exactly one unseen drug, and no pairs contain two unseen drugs.

Table 3: Cold-start partition of the test set, organized by the training-split membership of each drug in the pair.

Partition	Definition	n	% of test
warm-warm	both drugs in training	12,997	99.96%
warm-cold	one drug in training	5	0.04%
cold-cold	neither drug in training	0	0.00%
Total		13,002	100%

Two main issues arise from this distribution. The warm-cold partition is too small to provide reliable scores, with only 5 pairs falling below the 10-pair minimum used elsewhere in this work, and the cold-cold partition is empty. Per-partition metrics confirm that each model can be evaluated only in the warm-warm regime. Consequently, the headline numbers in Table 2 reflect performance when both drugs are familiar to the model, and it remains structurally impossible to determine whether any of these models can generalize to a novel drug.

The level of saturation is also a consequence of the splitting method rather than the data itself. Random pair-based splitting cannot produce drug-disjoint partitions when the negative sampler selects from the entire drug pool, because each drug will eventually appear in some training pair either as a positive or negative example. Implementing a drug-disjoint splitting strategy that removes whole drugs rather than pairs could resolve this, but it would reduce the training set size and alter the baselines that originally prompted this comparison. Section 6 explores this trade-off further.

The main deployment scenario affected by the inability to evaluate cold-start performance is screening new agents added after the initial benchmark. Current WHO TB and HIV guidelines include delamanid, pretomanid, and dolutegravir, but there is not enough test-pair coverage to classify them within the TB tiers in Section 5.5. The benchmark cannot answer whether a model trained on it can reliably identify interactions involving these drugs.

5.4. Class-Stratified Disagreement Between Architectures

Aggregate accuracy obscures class-specific differences in model errors. Throughout this subsection, an over-alerter denotes a model with high recall and low precision on the rare Minor class, and an under-alerter denotes the inverse pattern of high precision and low recall. Table 4 presents severity metrics for each class in the GNN+PK and MLP ensemble. The MLP is selected as the top-accuracy model and the GNN+PK as the best-calibrated, enabling a contrast between two architecturally distinct top performers.

Table 4: Per-class severity metrics for the GNN+PK and MLP ensemble on the fixed test set (n = 13,002). Precision, recall, and F1 scores are computed at the argmax over the four severity classes. AUROC and AUPRC are assessed in a class-vs-rest setup. Class supports show the distribution of the natural test set after negative sampling. 95% bootstrap confidence intervals for per-class P/R/F1 are included in Appendix A.3.

True class	**n**	**GNN+PK P**	**GNN+PK R**	**GNN+PK F1**	**MLP P**	**MLP R**	**MLP F1**
none	6,501	0.987	0.950	0.968	0.995	0.977	0.986
Minor	302	0.356	0.877	0.507	0.872	0.676	0.761
Moderate	4,679	0.887	0.749	0.812	0.917	0.970	0.943
Major	1,520	0.614	0.828	0.705	0.922	0.866	0.893

The Minor class exhibits opposite failure modes across the two architectures. GNN+PK has a recall of 0.877 but a precision of only 0.356, meaning it detects most true Minor cases but produces about two false positives for each correct detection. The MLP achieves a precision of 0.872 and a recall of 0.676, making its Minor predictions generally reliable, though it misses roughly one-third of actual Minor cases. Both models achieve similar F1 scores for this class through different mechanisms, and this difference is not reflected in their overall accuracy or macro-F1. A pairwise agreement analysis on severity correctness highlights the architectural pattern. Table 5 presents the four-cell agreement matrix comparing GNN+PK and MLP, with disagreements categorized by true severity class.

The two models disagree on 1,768 out of 13,002 test pairs, representing 13.6%. These disagreements mainly occur in the Moderate class, where the MLP outperforms with 1,063 wins to 26, roughly 41 to 1. In the Major class, the MLP wins 148 to 89, approximately 1.7 to 1, and in the "none" class, 244 to 68, about 3.6 to 1. The only exception is the Minor class, where GNN+PK outperforms, winning 64 to 3, roughly 21 to 1. The pattern indicates that the MLP outperforms in the Moderate and "none" classes, while GNN+PK has an advantage in the rare Minor class.

This pattern extends to the tree-based baseline models. GNN+PK surpasses XGB on Minor-class disagreements, winning 138 of 150 cases where the two models differ, with XGB correct in 1 and neither correct in the remaining 11. The MLP outperforms XGB on the 87 Minor-class disagreements between them, winning 77 to 1 with neither correct in 9. Across all six pairwise comparisons, the four models divide into two groups on the rare Minor class. GNN+PK alone employs an over-alerter strategy characterized by high recall and low precision, whereas MLP, RF, and XGB adopt an under-alerter strategy with high precision but lower recall. MLP, RF, and XGB each perform better than GNN+PK on Moderate because their under-alerter high-precision approach is effective for the dominant class, while GNN+PK excels on Minor due to its over-alerter high-recall pattern in the rare class.

Pairwise statistical tests confirm that this division is unlikely to be due to chance. The MLP surpasses GNN+PK by 0.093 in accuracy, with a bootstrap 95% confidence interval

Table 5: Pairwise agreement matrix comparing GNN+PK and MLP on the test set (n = 13,002). The top section shows four-cell agreement regarding severity correctness. The bottom section breaks down all disagreements by true severity class.

Outcome	**n**	**% of test**
Both correct	10,957	84.3%
GNN+PK only correct	247	1.9%
MLP only correct	1,458	11.2%
Both wrong	340	2.6%
Overall agreement	11,234	86.4%

True class	**n disagreements**	**GNN+PK wins**	**MLP wins**	**Neither**
none	332	68	244	20
Minor	78	64	3	11
Moderate	1,102	26	1,063	13
Major	256	89	148	19
Total	1,768	247	1,458	63

Note: Total by-class disagreements (1,768) surpass the combined number of "GNN+PK only correct" and "MLP only correct" pairs (247 + 1,458 = 1,705) because 63 of the 340 "Both wrong" pairs involve different incorrect predictions and are disagreements, while the remaining 277 share the same incorrect prediction and are counted as agreements.

of [0.087, 0.099] and a McNemar p-value below 10^{-6}. XGB outperforms GNN+PK by 0.028 in accuracy, with a confidence interval of [0.022, 0.035] and a McNemar p-value under 10^{-6}. These differences mainly stem from the under-alerter group's higher precision on the dominant Moderate class, where approximately 36% of test pairs are located. The strategies of over-alerters and under-alerters indicate distinct alert behaviors in practical deployment.

The clinical implications of these patterns bear directly on alert behavior. Severity classification in clinical decision support influences whether and how loudly alerts are issued. GNN+PK tends to over-predict Minor severity, generating false Minor flags from actual Moderate cases, with 380 such reclassifications in the test set. This produces insufficient warnings for dose-dependent Moderate interactions alongside excessive alerts for cases that are not genuinely Minor. The MLP under-predicts Minor severity, allowing those cases to pass without alerts. Clinicians using either model would encounter different error patterns despite similar overall accuracy, and the implications for alert fatigue and patient safety differ in ways the benchmark's aggregate metrics do not reveal.

5.5. TB-Tier Performance

Performance on TB-relevant pairs diverges from the full-test results, reinforcing the three audit findings. Table 6 presents tier-based metrics for all four main models.

All models perform worse on TB-any pairs than on the complete test set, with accuracy decreasing by 6 to 14 percentage points and similar drops in macro-F1. The MLP shows the best overall performance across all tiers, but its lead over the next-best model narrows

Table 6: Performance comparison across TB drug tiers. Rows represent the four tiers described in Section 4.1, along with the union (TB-any) and the complete test set for reference. Columns display accuracy, macro-F1, and expected calibration error (ECE) for each model.

Tier	n	RF Acc/F1/ECE	MLP Acc/F1/ECE	XGB Acc/F1/ECE	GNN+PK Acc/F1/ECE
Full test	13,002	0.849 / 0.707 / 0.017	0.955 / 0.896 / 0.011	0.890 / 0.781 / 0.014	0.862 / 0.748 / 0.008
TB (any)	313	0.754 / 0.530 / 0.078	0.898 / 0.843 / 0.043	0.789 / 0.652 / 0.055	0.722 / 0.692 / 0.045
First-line	32	0.688 / 0.410 / 0.115	0.812 / 0.641 / 0.110	0.750 / 0.477 / 0.146	0.750 / 0.663 / 0.091
Second-line	82	0.756 / 0.444 / 0.154	0.951 / 0.972 / 0.052	0.842 / 0.803 / 0.126	0.756 / 0.713 / 0.083
ARV co-admin	72	0.736 / 0.604 / 0.086	0.861 / 0.706 / 0.099	0.750 / 0.550 / 0.034	0.667 / 0.688 / 0.135
Comedications	129	0.783 / 0.563 / 0.091	0.907 / 0.863 / 0.060	0.791 / 0.656 / 0.059	0.721 / 0.685 / 0.107

in the rifampin-heavy tiers. The First-line and ARV co-administration tiers are especially challenging for all models. GNN+PK's accuracy falls to 0.667 on ARV co-administration, and XGB achieves only 0.550 macro-F1 in that tier. These tiers involve rifampin-induced CYP induction interactions, which are clinically important, and all models perform worse on these key cases. Calibration declines more noticeably in smaller tiers than overall accuracy suggests. GNN+PK's ECE increases from 0.008 on the full dataset to 0.091 on first-line drug pairs and 0.135 on ARV co-administered pairs, indicating that a well-calibrated overall model may still perform poorly within the specific subgroups most relevant to clinicians.

This tier-stratified view consolidates the three audit findings. Name-resolution gaps determine which TB drugs belong to the cohort. Partition saturation limits evaluation of agents not yet incorporated into the benchmark. Class-stratified disagreement shapes the type of alert error a clinician encounters when querying an underrepresented tier. These issues are interconnected, as the same drugs are affected by all three findings concurrently. Section 6 discusses the implications.

6. Discussion

This audit's insights extend beyond identifying technical bias patterns to evaluating what a clinical decision support tool for DDI screening can reliably achieve, especially for key populations involved in global tuberculosis and HIV care. As Section 5.5 shows, a model's strong overall calibration can mask tier-level error more than ten times larger on the small clinical subgroups that matter most for TB-HIV prescribing. The difference between overall performance metrics and tier-specific accuracy affects clinicians, regulators, and patients, but only developers can reveal this discrepancy.

This highlights a broader issue in ensuring accountability in clinical AI. A clinical decision support system evaluated against this benchmark would likely be considered a moderate-risk Software as a Medical Device under FDA regulations (U.S. Food and Drug Administration, 2017), necessitating clinician oversight rather than full automation. This oversight assumes clinicians can assess whether the system's performance suits the specific patient group. Without features such as name-resolution diagnostics, partition transparency, and class-stratified reporting, clinicians must rely on performance data they cannot independently verify, placing responsibility for validation on end-users rather than developers. The drugs most impacted by this shift are those linked to WHO INN names, which are central to global TB and HIV treatments. Validation gaps in such benchmarks tend to harm patient populations with less infrastructure to detect tool failures.

The audit highlights an often overlooked aspect of model selection. The MLP ensemble outperforms GNN+PK in accuracy by 9.3 percentage points and offers faster training times and greater ease of deployment. The GNN+PK shows consistent benefits only in calibration (0.008 versus 0.011) and Minor-class detection, both important for alert behavior but with narrow margins. In deployment scenarios where the choice is between a well-tuned descriptor-based model and a more complex graph-based model that demands greater computation and careful pretraining, the descriptor-based model may be the more dependable option. These findings apply equally to both architectures, suggesting that increased complexity does not necessarily enhance benchmark validation.

Three concrete adjustments to the standard DDI benchmark approach follow from this work. Name-resolution diagnostics should be included alongside aggregate cohort statistics, with explicit reporting of which drugs were excluded and the reasons. Partition design should match the deployment scenario, since drug-disjoint splits are necessary when screening newer agents. Class-stratified reporting should accompany aggregate metrics whenever clinical outcomes vary by severity. Whether similar patterns influence benchmarks in related clinical machine learning areas is an empirical question for future studies.

Limitations The audit evaluates a single DDI benchmark, and cross-benchmark validation is needed before broader application. Mechanism labels are derived through regex pattern matching against DrugBank's free-text descriptions, and a portion of positive pairs lack mechanism keywords and are excluded from mechanism training. The reported mechanism-head performance therefore reflects the quality of regex-derived labels rather than true pharmacological mechanisms.

Indirect label leakage in the pharmacokinetic features was identified and corrected during this work. Per-mechanism contingency analysis on the training pairs showed CYP-inducer flags carrying a $2.5\times$ lift on CYP induction labels and CYP-inhibitor flags carrying a $2.2\times$ lift on CYP inhibition labels, confirming that the affected columns encoded downstream mechanism information rather than independent chemistry. The corrected pipeline removes 20 of the original 30 PK columns, including all CYP induction and inhibition flags, and all Section 5 metrics use the leakage-corrected vector. Researchers building similar benchmarks should audit their PK features for the same pattern, since training-set lift ratios above $2\times$ for any feature against a label warrant scrutiny.

The TB analysis includes only 18 drugs with sufficient test coverage. The molecular graph encoder uses a moderate-depth GATv2 architecture without pretraining. The encoder applies one-hot atom and bond featurization (49 atom dimensions and 14 bond dimensions) standard in graph-based molecular property prediction, so the comparison against fingerprint-based baselines reflects representation quality rather than featurization shortcuts. Variants with pretrained encoders may shift accuracy and AUROC values, but encoder architecture choice is unlikely to affect the audit's three main conclusions.

References

Shaked Brody, Uri Alon, and Eran Yahav. How attentive are graph attention networks? In *International Conference on Learning Representations*, 2022.

William J. Burman, Keith Gallicano, and Charles Peloquin. Comparative pharmacokinetics and pharmacodynamics of the rifamycin antibacterials. *Clinical Pharmacokinetics*, 40(5): 327–341, 2001.

Mauren Felisberto, Graziella dos Santos Lima, Ivan Claudio Celuppi, et al. Override rate of drug-drug interaction alerts in clinical decision support systems: A brief systematic review and meta-analysis. *Health Informatics Journal*, 30(2), 2024. doi: 10.1177/14604582241263242.

Kexin Huang, Tianfan Fu, Wenhao Gao, Yue Zhao, Yusuf Roohani, Jure Leskovec, Connor W. Coley, Cao Xiao, Jimeng Sun, and Marinka Zitnik. Therapeutics Data Commons: Machine learning datasets and tasks for drug discovery and development. In *Proceedings of Neural Information Processing Systems Track on Datasets and Benchmarks*, 2021.

Roman Joeres, David B. Blumenthal, and Olga V. Kalinina. Data splitting to avoid information leakage with DataSAIL. *Nature Communications*, 16:3337, 2025. doi: 10.1038/s41467-025-58606-8.

Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023.

Zhe Liu, Xiao-Nan Wang, Hui Yu, Jian-Yu Shi, and Wei-Ming Dong. Predict multi-type drug-drug interactions in cold start scenario. *BMC Bioinformatics*, 23:75, 2022.

Harry L. Morgan. The generation of a unique machine description for chemical structures: A technique developed at Chemical Abstracts Service. *Journal of Chemical Documentation*, 5(2):107–113, 1965.

Bret Nestor, Matthew B. A. McDermott, Willie Boag, Gabriela Berner, Tristan Naumann, Michael C. Hughes, Anna Goldenberg, and Marzyeh Ghassemi. Feature robustness in non-stationary health records: Caveats to deployable model performance in common clinical machine learning tasks. In *Proceedings of the 4th Machine Learning for Healthcare Conference*, volume 106 of *PMLR*, pages 1–23, 2019.

Mikko Niemi, Janne T. Backman, Martin F. Fromm, Pertti J. Neuvonen, and Kari T. Kivistö. Pharmacokinetic interactions with rifampicin: Clinical relevance. *Clinical Pharmacokinetics*, 42(9):819–850, 2003.

Arnold K. Nyamabo, Hui Yu, and Jian-Yu Shi. SSI-DDI: Substructure-substructure interactions for drug-drug interaction prediction. *Briefings in Bioinformatics*, 22(6):bbab133, 2021.

Arnold K. Nyamabo, Hui Yu, Zhe Liu, and Jian-Yu Shi. Drug-drug interaction prediction with learnable size-adaptive molecular substructures. *Briefings in Bioinformatics*, 23(1): bbab441, 2022. doi: 10.1093/bib/bbab441.

David Rogers and Mathew Hahn. Extended-connectivity fingerprints. *Journal of Chemical Information and Modeling*, 50(5):742–754, 2010.

Jae Yong Ryu, Hyun Uk Kim, and Sang Yup Lee. Deep learning improves prediction of drug-drug and drug-food interactions. *Proceedings of the National Academy of Sciences*, 115(18):E4304–E4311, 2018.

Zhiyuan Shen, Ming Zhou, Yanbin Zhang, and Qingsong Yao. Benchmarking drug-drug interaction prediction methods: A perspective of distribution changes. *Bioinformatics*, 41(11):btaf569, 2025. doi: 10.1093/bioinformatics/btaf569.

U.S. Food and Drug Administration. Software as a Medical Device (SaMD): Clinical evaluation. Guidance for Industry and Food and Drug Administration Staff, 2017.

Yunsheng Wang, Yingqing Min, Xu Chen, and Jure Wu. Multi-view graph contrastive representation learning for drug-drug interaction prediction. In *Proceedings of the Web Conference 2021*, pages 2921–2933, 2021.

David S. Wishart, Yannick D. Feunang, An C. Guo, et al. DrugBank 5.0: A major update to the DrugBank database for 2018. *Nucleic Acids Research*, 46(D1):D1074–D1082, 2018.

Adrian Wong, Erkin Otles, John P. Donnelly, et al. External validation of a widely implemented proprietary sepsis prediction model in hospitalized patients. *JAMA Internal Medicine*, 181(8):1065–1070, 2021.

Guoli Xiong, Zhijiang Yang, Jiacong Yi, et al. DDInter: An online drug-drug interaction database towards improving clinical decision-making and patient safety. *Nucleic Acids Research*, 50(D1):D1200–D1207, 2022.

Keyulu Xu, Chengtao Li, Yonglong Tian, Tomohiro Sonobe, Ken-ichi Kawarabayashi, and Stefanie Jegelka. Representation learning on graphs with jumping knowledge networks. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *PMLR*, pages 5453–5462, 2018.

Marinka Zitnik, Monica Agrawal, and Jure Leskovec. Modeling polypharmacy side effects with graph convolutional networks. *Bioinformatics*, 34(13):i457–i466, 2018.

Appendix A. Hyperparameters

A.1. Baseline Models

The Random Forest, MLP, and XGBoost classifiers were trained with the hyperparameters listed in Table 7. All baselines use 5-fold stratified cross-validation on the training split with the same random seed (42).

Table 7: Hyperparameters for baseline severity and mechanism classifiers.

Hyperparameter	RF (sev)	MLP (sev)	XGB (sev)	RF (mech)	MLP (mech)	XGB (mech)
n_estimators	300	N/A	500	200	N/A	300
max_depth	20	N/A	6	15	N/A	6
hidden_layers	N/A	(512, 256)	N/A	N/A	(256, 128)	N/A
max_iter / epochs	N/A	100	N/A	N/A	80	N/A
learning_rate	N/A	N/A	0.05	N/A	N/A	0.05
subsample	N/A	N/A	0.8	N/A	N/A	1.0
colsample_bytree	N/A	N/A	0.6	N/A	N/A	1.0
early_stopping	N/A	yes	N/A	N/A	yes	N/A
validation_fraction	N/A	0.1	N/A	N/A	0.1	N/A
calibration	isotonic, cv=3	none	isotonic, cv=3	isotonic OvR, cv=3	OvR	isotonic OvR, cv=3

Random forests and XGBoost classifiers are wrapped in scikit-learn's `CalibratedClassifierCV` with isotonic regression for severity prediction. Mechanism heads use `OneVsRestClassifier` wrapping for multi-label binary prediction.

A.2. GNN Architecture

The shared encoder uses three GATv2 convolutional layers with 4 attention heads per layer, hidden dimension 128, edge feature dimension 14, and dropout 0.2 with edge dropout 0.15. Jumping knowledge aggregates layer outputs in max mode. The pooling stage concatenates global mean and global max pooling, producing a 256-dimensional molecule embedding.

The pharmacokinetic branch is a two-layer MLP mapping the 10-dimensional input vector to a 64-dimensional embedding with ReLU activation and dropout 0.2.

The pair classifier trunk takes the 320-dimensional per-drug representation, builds a 1,280-dimensional interaction vector via concatenation of $[\mathbf{h}_a; \mathbf{h}_b; \mathbf{h}_a \odot \mathbf{h}_b; |\mathbf{h}_a - \mathbf{h}_b|]$, and processes it through two linear layers (512, 256 dimensions) with ReLU and dropout 0.2 between heads. The severity head outputs four logits, the mechanism head outputs seven.

A.3. Training

Training was conducted on an Apple M4 Max with 128 GB RAM using the PyTorch MPS backend.

Appendix B Per-class Severity Metrics with Bootstrap Confidence Intervals

Confidence intervals derive from 1,000 bootstrap resamples of the test set with 95% percentile bounds. All values are reported on the fixed test set of 13,002 drug pairs.

Table 8: GNN training settings.

Setting	**Value**
Optimizer	Adam
Learning rate	10^{-3}
Weight decay	10^{-5}
Batch size	512
Max epochs	80
Patience	15
Gradient clip norm	1.0
LR schedule	Cosine annealing
Cosine $\eta_{\min}$	10^{-6}
Mechanism loss weight (multi-task)	0.5
Mechanism loss weight (single-task)	0.0
Negative sampling ratio	1:1
Negative oversample factor	2
Random seed	42
Train / Validation / Test split	80 / 10 / 10

B.1 GNN+PK

Class	n	**Precision**	**Recall**	**F1**
none	6,501	0.987 [0.985, 0.990]	0.950 [0.945, 0.956]	0.968 [0.965, 0.972]
Minor	302	0.355 [0.323, 0.392]	0.877 [0.838, 0.912]	0.505 [0.470, 0.544]
Moderate	4,679	0.887 [0.877, 0.897]	0.749 [0.736, 0.760]	0.812 [0.803, 0.820]
Major	1,520	0.614 [0.593, 0.635]	0.827 [0.808, 0.847]	0.705 [0.687, 0.721]

B.2 MLP (5-fold ensemble)

Class	n	**Precision**	**Recall**	**F1**
none	6,501	0.995 [0.993, 0.997]	0.978 [0.974, 0.981]	0.986 [0.984, 0.988]
Minor	302	0.871 [0.828, 0.910]	0.675 [0.621, 0.728]	0.760 [0.718, 0.800]
Moderate	4,679	0.917 [0.909, 0.924]	0.970 [0.965, 0.975]	0.943 [0.938, 0.947]
Major	1,520	0.921 [0.907, 0.935]	0.866 [0.848, 0.882]	0.893 [0.881, 0.904]

B.3 RF (5-fold ensemble)

Class	n	**Precision**	**Recall**	**F1**
none	6,501	0.914 [0.907, 0.920]	0.942 [0.936, 0.948]	0.928 [0.923, 0.932]
Minor	302	0.859 [0.793, 0.915]	0.340 [0.287, 0.395]	0.487 [0.423, 0.544]
Moderate	4,679	0.779 [0.768, 0.790]	0.873 [0.863, 0.883]	0.823 [0.816, 0.832]
Major	1,520	0.774 [0.747, 0.802]	0.478 [0.454, 0.502]	0.591 [0.568, 0.614]

B.4 XGB (5-fold ensemble)

Class	n	Precision	Recall	F1
none	6,501	0.956 [0.951, 0.961]	0.963 [0.959, 0.968]	0.960 [0.957, 0.963]
Minor	302	0.806 [0.745, 0.861]	0.482 [0.424, 0.538]	0.603 [0.551, 0.650]
Moderate	4,679	0.832 [0.822, 0.842]	0.901 [0.893, 0.910]	0.865 [0.858, 0.873]
Major	1,520	0.787 [0.764, 0.807]	0.622 [0.599, 0.644]	0.694 [0.676, 0.714]

The Minor class confidence intervals quantify the architecture-specific failure modes discussed in Section 5.4. The GNN+PK precision interval [0.323, 0.392] does not overlap with the MLP, RF, or XGB precision intervals. The GNN+PK recall interval [0.838, 0.912] does not overlap with the RF or XGB recall intervals. The two failure modes (graph-based over-flagging, descriptor-based under-flagging) are statistically distinct rather than within-noise differences.